

No Finite Separable Representation at Three Prime Generators

Kernel rank and a tail criterion for Erdős Problem #269

WILL COOK*

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ABSTRACT

For three distinct primes, the reciprocal running-LCM kernel has nonsingular minors of every order, with the same indices in every third-coordinate layer. Row and column rescaling reduces the proof to a matrix whose entries are 1 or the reciprocal of the third prime. Thus no finite sum of products separates one exponent from the other two. Separately, for the repeated $\{2, 3, 5\}$ series we prove a tail recurrence and a residue criterion equivalent to irrationality. The unresolved step is to prove its inequalities for every positive multiplier coprime to 30 and arbitrarily late starts.

1 Introduction

For a finite prime set P , let $\mathcal{S}_P$ consist of the positive integers whose prime factors lie in P , including 1. We use $\mathbb{N} = \{0, 1, 2, \dots\}$ for exponent indices. For $x \geq 1$, put

$$L(x) = \text{lcm}\{u \in \mathcal{S}_P : u \leq x\}, \quad \mathcal{R}_P = \sum_{u \in \mathcal{S}_P} L(u)^{-1}.$$

Erdős asked whether $\mathcal{R}_P$ is irrational when $|P| \geq 2$ [1, p. 65][2, p. 106]. For $P = \{p, q, r\}$, write

$$H(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor}, \quad K(i, j, k) = H(p^i q^j r^k)^{-1}.$$

Prime-power divisibility gives $L = H$ (Proposition 4.1). Fan used this identity to separate the two-prime kernel [7]. With three primes, even a finite sum of separated products is impossible. This is a statement about the kernel, not the arithmetic nature of its sum.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

2 Nonsingular minors of every order

Theorem 2.1 (no finite separation of the kernel). *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 0$ there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j, k)$, $0 \leq \ell < d$, satisfying

$$K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k) \quad \text{for all } i, j, k.$$

Lean: [paper uniform rank and nonseparation](#).

Proof. For fixed k , we rescale rows and columns to obtain a matrix independent of k . Put $c = r^{-1}$, $x_i = \{i \log_r p\}$ and $y_j = \{j \log_r q\}$, where braces denote fractional parts. Splitting the floor exponents gives

$$K(i, j, k) = U_i(k)^{-1} C_{ij} V_j(k)^{-1}, \quad C_{ij} = c^{\mathbf{1}_{\{x_i + y_j \geq 1\}}},$$

where

$$U_i(k) = p^i q^{\lfloor \log_q(p^i r^k) \rfloor} r^{k + \lfloor \log_r p^i \rfloor},$$

$$V_j(k) = q^j p^{\lfloor \log_p(q^j r^k) \rfloor} r^{\lfloor \log_r q^j \rfloor}.$$

The positive factors do not affect whether a minor vanishes; C records only whether $x_i + y_j \geq 1$. Distinct primes make $\log_r p$ and $\log_r q$ irrational, so each fractional-part orbit is dense in $[0, 1]$. The row and column indices are independent choices; no density of a single orbit of pairs is required. For $n \geq 1$, choose distinct rows with $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$. We choose each column so that its entries change from 1 to c at a different selected row. Writing $s_b = 1 - y_{J(b)}$, density of (y_j) lets us choose

$$s_0 \in (0, x_{I(0)}), \quad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals for the s_b are disjoint, so the column indices are distinct. The strict inequalities also avoid the threshold itself, and give $x_{I(a)} + y_{J(b)} \geq 1$ exactly when $b \leq a$. Thus multiplying row a by $U_{I(a)}(k)$ and column b by $V_{J(b)}(k)$ reduces the selected kernel minor to

$$T_n(c) = \begin{pmatrix} c & 1 & \cdots & 1 \\ c & c & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \cdots & c \end{pmatrix}, \quad \det T_n(c) = c(c-1)^{n-1}.$$

Indeed, subtracting each preceding row from the next, working upwards from the last row, leaves diagonal entries $c, c-1, \dots, c-1$. The original determinant is therefore

$$c(c-1)^{n-1} \prod_{a < n} U_{I(a)}(k)^{-1} \prod_{b < n} V_{J(b)}(k)^{-1} \neq 0.$$

The indices were chosen from C , independently of k . For $n = 0$ the empty determinant equals 1.

For the final assertion, suppose a separation with d summands existed and fix any k . On the rows $I(0), \dots, I(d)$ and columns $J(0), \dots, J(d)$ its matrix would be the product of the $(d+1) \times d$ matrix $(f_\ell(I(a)))_{a,\ell}$ and the $d \times (d+1)$ matrix $(G_\ell(J(b), k))_{\ell,b}$. Its rank would be at most d , so its determinant would vanish, contrary to the minor just constructed. $\square$

No continuity or boundedness is assumed for the separated factors.

Corollary 2.2 (the same minors modulo integers coprime to 30). *For $(p, q, r) = (2, 3, 5)$ and every $n \geq 1$, there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$, chosen independently of B and k , such that for every $B \geq 2$ coprime to 30 and every $k \geq 0$, the selected $n \times n$ kernel matrix has unit determinant over $\mathbb{Z}/B\mathbb{Z}$, with each reciprocal prime power interpreted by its modular inverse.*

Lean: [admissible modular minors](#).

Proof. Choose the maps from Theorem 2.1. Every row and column factor is a unit modulo B . The normalised determinant is $5^{-1}(-4/5)^{n-1}$, also a unit. $\square$

The modulus may be composite, for example 49 or 77. Coprimality with 30 makes both the reciprocal entries and the determinant units: the latter introduces only a power of 4 in its numerator.

Example 2.3. For $(p, q, r) = (2, 3, 5)$, the leading two-by-two determinant is $-1/15$: its four entries are $1, 1/6, 1/2, 1/60$, so it equals $1/60 - 1/12$.

The indices must be selected. The leading 4×4 block at $\{2, 3, 5\}$ is singular:

$$K(3, j, 0) = \frac{1}{120} K(0, j, 0) \quad (0 \leq j < 4).$$

These equalities follow by evaluating the height at 3^j and $8 \cdot 3^j$. But at $j = 4$, $K(3, 4, 0) - K(0, 4, 0)/120 = -1/19440000$. Thus the leading minors do not supply the arbitrary-order theorem.

Proposition 2.4 (rank of a matrix of threshold columns). *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq k \leq m$ write v_k for the length- m column with a 1 in each of the first k coordinates and c thereafter. A matrix whose distinct columns are v_k for k in a nonempty set E has rank $|E| - \mathbf{1}_{\{0, m\} \subseteq E}$.*

Lean: [rank cut matrix](#).

The correction accounts for the two constant columns: $v_0 = cv_m$. There is no other dependence among distinct threshold columns.

Proof. List $E = \{k_1 < \dots < k_t\}$. The $t-1$ differences $v_{k_{j+1}} - v_{k_j} = (1-c)\mathbf{1}_{\{k_j, \dots, k_{j+1}-1\}}$ have disjoint nonempty supports, hence are linearly independent. If $k_1 > 0$ or $k_t < m$, their union misses a coordinate where v_{k_1} is nonzero, so the rank is t . If $k_1 = 0$ and $k_t = m$, then $v_0 = c\mathbf{1}$ lies in the span of the differences (their sum is $(1-c)\mathbf{1}$), so the rank is $t-1$. $\square$

The restrictions $c \neq 0, 1$ exclude a zero column or identical columns; they hold for $c = 1/r$ over $\mathbb{Q}$. In one fixed layer k , ordering the sampled row phases puts every column in the stated form. The formula therefore determines the rank of each nonempty rectangular sample. For fixed row and column indices this rank is independent of k . The companion, Section 4, gives the exact rational comparisons.

Exact rank is not an approximation obstruction. Truncating the original kernel to $i < N$ gives a sum of N separated terms. The bound $H(x) > x^3/(pqr)$ makes the truncation error tend to zero both uniformly and in $\ell^1(\mathbb{N}^3)$. By contrast, the rescaled matrix in the rank proof has uniform distance $(1 - 1/r)/2$ from the matrices of finite rank. Rescaling preserves exact rank but removes the decay responsible for the first approximation. The companion proves both assertions, including the distinction between finite restrictions and the full infinite matrix, in Section 4.1.

3 The two-prime comparison

Fan's post of 26 June 2026 gives the repeated two-prime factorisation, its quadratic expression in one Hecke–Mahler value and its transcendence conclusion [7]. Counting the contributions at powers of each prime also gives the affine formula for the distinct-height sum below. The calculation fixes the normalisations and distinguishes the two sums.

Theorem 3.1 (both two-prime sums). *Let $p < q$ be distinct primes. Put $\theta = \log p / \log q$ and $A = \sum_{n \geq 0} p^{-n} q^{-\lfloor n\theta \rfloor}$. Let $\mathcal{R}_{p,q}$ sum the reciprocal running LCM at every positive $\{p, q\}$ -smooth integer, and let $\mathcal{D}_{p,q}$ count each distinct running LCM once. Then*

$$\mathcal{D}_{p,q} = \frac{(q-p)A + p}{q-1}, \quad \mathcal{R}_{p,q} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \quad (1)$$

Both numbers are transcendental.

Lean: [two prime affine and quadratic](#), [two prime sums transcendental](#), [running lcm eq two prime height](#), [two prime height smooth](#), and 9 further declarations. Conditional on the transcendence theorem of Bugeaud and Laurent; see the [coverage section of the companion record](#).

Proof. Set $x = 1/p$, $y = 1/q$, $m_n = \lfloor n\theta \rfloor$ and $\delta_n = m_{n+1} - m_n$. Unique factorisation makes θ irrational, and $0 < \theta < 1$ gives $\delta_n \in \{0, 1\}$. The initial value and the positive powers of p contribute $A = \sum_{n \geq 0} x^n y^{m_n}$. There is one q -power strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, so the positive powers of q contribute

$$B = \sum_{n \geq 0} \delta_n x^n y^{m_n+1}, \quad \mathcal{D}_{p,q} = A + B.$$

Since $0 < x, y < 1$, these series converge absolutely. The identity $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$ and an index shift in A give $A - 1 - xA = x(y - 1)B/y$. Therefore

$$B = \frac{p - (p-1)A}{q-1}.$$

At $p^i q^j$ the height is $p^{i+\lfloor j/\theta \rfloor} q^{j+m_i}$, hence

$$\mathcal{R}_{p,q} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor} = A(1 + B).$$

For the last equality, each $j \geq 1$ corresponds to $n = \lfloor j/\theta \rfloor$ with $m_n = j - 1$ and $\delta_n = 1$. These identities prove (1).

For the Hecke–Mahler series $F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{m_n} x^n y^k$, geometric summation gives

$$A = \frac{1}{1-x} - \frac{1-y}{y} F_\theta(x, y). \quad (2)$$

Bugeaud and Laurent’s theorem [5, Theorem 1.1] applies with intercept $\rho = 0$, $\beta = x$ and $\alpha = y$: the slope θ is irrational and lies in $(0, 1)$, x and y are nonzero algebraic numbers, $|x| < 1$ and $|xy^\theta| = p^{-2} < 1$. Hence $F_\theta(x, y)$ is transcendental; this case $\rho = 0$ is due to Loxton and van der Poorten [6, Theorem 8, p. 40]. Thus A is transcendental. The displayed affine and quadratic polynomials are nonconstant, so an algebraic value of either would force A to be algebraic. $\square$

The same calculation applies to the monoid generated by any coprime integers $1 < p < q$, such as 4, 9, but need not describe all integers supported on their prime factors. The companion’s Section 3 gives this extension and the failure for 4, 8.

The three-prime tail argument starts again from the running LCM, retaining every multiplicity; it does not use the rank theorem. The distinct-height assertion is a separate historical statement [3].

4 The recurrence for the repeated sum

Proposition 4.1 (the running least common multiple). *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then the running LCM equals the three-prime height: $L(x) = H(x)$.*

Lean: [running lcm real cutoff exact](#).

Proof. Every smooth $n \leq x$ has prime exponents bounded by the corresponding integer logarithms, so $n \mid H(x)$. Conversely the three maximal pure powers occur among those smooth numbers; their product divides the running LCM because they are pairwise coprime. $\square$

The inequalities $x/p < p^{\lfloor \log_p x \rfloor} \leq x$ for each of the three primes also give

$$x^3 / (pqr) < H(x) \leq x^3. \quad (3)$$

Proposition 4.2 (cells and jumps). *The running LCM is constant when the three integer logarithms are constant. A jump in exactly one logarithm multiplies it by the corresponding prime. The first n positive powers of each prime, together with 1, form $3n + 1$ distinct points.*

Lean: [real prefix lcm eq of same log cell](#), [real prefix lcm jump first](#), [real prefix lcm jump second](#), [real prefix lcm jump third](#), and 1 further declaration in the [coverage section of the companion record](#).

Proof. The first two claims follow from the height formula. Positive powers of different primes cannot coincide, by unique factorisation, and none is 1. $\square$

For $(p, q, r) = (2, 3, 5)$, the smooth numbers 5 and 6 have the same running LCM, 60. They contribute $2/60$ to the repeated sum, whereas the height 60 contributes only $1/60$ to the distinct-height sum. The next identity retains this multiplicity when equal heights are grouped.

Proposition 4.3 (grouping terms with the same height). *For a finite exponent box $\mathcal{B}$, set $F(H) = \{(i, j, k) \in \mathcal{B} : H(p^i q^j r^k) = H\}$. Then*

$$\sum_{(i,j,k) \in \mathcal{B}} K(i, j, k) = \sum_H \frac{\#F(H)}{H}. \quad (4)$$

Lean: [finite smooth kernel sum grouped by height](#).

Proof. Each term in $F(H)$ equals $1/H$. The multiplicities remain present in every subsequent infinite sum. $\square$

From now on let $P = \{2, 3, 5\}$ and $S = \mathcal{R}_P$. For $a \geq 0$ define

$$s_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{1}{H(x)}, \quad T_a = \sum_{j \geq 0} s_{a+j}, \quad h_a = \frac{H(2^a)}{2}, \quad X_a = h_a T_a.$$

The half-height h_a clears the finite prefix: for $a \geq 1$ and smooth $x < 2^a$, the exponent of 2 in $H(x)$ is at most $a - 1$, so $H(x) \mid H(2^a)/2$. This is why we omit the final factor 2. Here $h_0 = 1/2$; only for $a \geq 1$ is h_a an integer. Define

$$b_a = \frac{H(2^{a+1})}{H(2^a)}, \quad m_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{H(2^{a+1})}{2H(x)}. \quad (5)$$

Lemma 4.4 (integer coefficients and four possible bases). *For every $a \geq 0$, m_a is a positive integer and $b_a \in \{2, 6, 10, 30\}$. The word “numerator” does not impose the positional-digit restriction $m_a < b_a$; that restriction need not hold.*

Lean: [dyadic alphabet whole](#).

Proof. If $x < 2^{a+1}$, the exponent of 2 in $H(x)$ is at most a , while the other exponents are bounded by those of $H(2^{a+1})$. Hence $2H(x) \mid H(2^{a+1})$. Each summand is an integer, and the shell contains 2^a . Between consecutive powers of 2 there is at most one power of 3 and at most one power of 5: successive powers of either odd prime have ratio greater than two. These factors, when present, and the factor 2 at the right endpoint give the four possible bases. $\square$

In the shell $[2, 4)$ the terms $1/2$ and $1/6$, with $h_2 = b_1 = 6$, give $m_1 = 4$ and the recurrence $X_2 = 6X_1 - 4$. The shell $[16, 32)$ shows why these numerators are not positional digits: $m_4 = 65 > b_4 = 30$. The companion computes the carried digits separately.

Proposition 4.5 (the tail recurrence and a quadratic bound). *The series defining S, T_a converge. For every $a \geq 0$,*

$$X_{a+1} = b_a X_a - m_a, \quad 0 < X_a \leq \frac{8640}{343}(a+1)^2 < 90(a+1)^2.$$

For every integer $B \geq 1$, either some BX_a is integral and all later states are integral, or $\text{dist}(BX_a, \mathbb{Z}) \geq 1/31$ at arbitrarily large indices.

Lean: [short actual orbit](#).

Proof. For each pair of exponents of 3 and 5, at most one exponent of 2 places the smooth integer in $[2^a, 2^{a+1})$. The two odd-prime exponents lie between 0 and a , giving at most $(a+1)^2$ terms. By (3), $s_a \leq 30(a+1)^2/8^a$. Since $h_a \leq 8^a/2$ and $a+j+1 \leq (a+1)(j+1)$ for $a, j \geq 0$, summing the later shells gives

$$X_a \leq 15(a+1)^2 \sum_{j \geq 0} \frac{(j+1)^2}{8^j} = \frac{8640}{343}(a+1)^2.$$

This proves convergence and the bound. Splitting the first shell gives the recurrence because $m_a = h_{a+1}s_a$ and $h_{a+1} = b_a h_a$. If all sufficiently late distances are strictly below $1/31$, write $BX_a = z_a + e_a$ with $z_a \in \mathbb{Z}$ and $|e_a| < 1/31$. Since Bm_a is integral, $e_{a+1} - b_a e_a$ is an integer of absolute value less than $(1 + b_a)/31 \leq 1$, so $e_{a+1} = b_a e_a$. The lower bound $b_a \geq 2$ forces this bounded error to vanish; integrality then propagates by the recurrence. $\square$

The identity $S/2 = \sum_{a \geq 0} m_a / (b_0 \cdots b_a)$ places the recurrence in the setting of Cantor series. The classical criteria of Erdős–Straus [8, Theorem 2.1] and Hančl–Tijdeman [9, Theorem 3.1] require $m_a / (b_{a-1} b_a) \rightarrow 0$. That hypothesis fails here: the numerators grow quadratically while the bases are bounded. The companion’s Section 6 proves the growth assertion and gives the precise carry conditions. The argument below instead uses the recurrence and the proved bound $90B(a+1)^2$ directly.

Theorem 4.6 (rationality gives positive integer tails). *If $S = A/D$ in lowest terms, where $D = 2^u 3^v 5^w B$ and $\gcd(B, 30) = 1$, then for every $a \geq a_0 = u + 1 + 2v + 3w$,*

$$d_a = BX_a \in \mathbb{Z}_{>0}, \quad d_{a+1} = b_a d_a - Bm_a, \quad d_a \leq 90B(a+1)^2.$$

Lean: [short fixed split bridge](#).

Proof. For $a \geq 1$, every $H(x)$ with $x < 2^a$ divides h_a . Subtracting the finite prefix therefore gives

$$X_a = h_a S - \sum_{\substack{x \text{ smooth} \\ x < 2^a}} \frac{h_a}{H(x)}, \quad \sum_{\substack{x \text{ smooth} \\ x < 2^a}} \frac{h_a}{H(x)} \in \mathbb{Z}. \quad (6)$$

The exponent of 2 in h_a is $a - 1$. Moreover $a \geq 2v$ gives $2^a \geq 3^v$, and $a \geq 3w$ gives $2^a \geq 5^w$. Thus $2^u 3^v 5^w \mid h_a$ after the stated onset, and BX_a differs from $h_a A / (2^u 3^v 5^w)$ by an integer and is therefore integral. Positivity, the recurrence and the bound follow from Proposition 4.5. $\square$

Every positive denominator splits in the required way: B is what remains after removing all factors 2, 3 and 5. For example, a hypothetical reduced denominator $2^3 3^2 5 \cdot 7$ would give $B = 7$; the next result gives the first integral $7X_a$ at $a = 4$, whereas the sufficient bound above gives $a_0 = 11$.

Corollary 4.7 (the first index at which the denominator clears). *Under the same lowest-terms hypothesis, put $M = 2^u 3^v 5^w$ and let den denote the positive reduced denominator. For $a \geq 1$,*

$$\text{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Consequently BX_a is integral exactly when $2^a \geq \max(2^{u+1}, 3^v, 5^w)$. The first such a can be found by integer comparisons, without logarithmic rounding.

Lean: [exact denominators and minimal clearing](#), [scaled state is integer iff first clearing index \$le\$](#) , [clearing condition iff max](#).

Proof. By (6), BX_a differs from $h_a A/M$ by an integer. Since $\gcd(A, M) = 1$, reducing that fraction gives the displayed denominator. The three divisibility conditions for h_a are precisely the three inequalities stated above. $\square$

5 A window test and the remaining arithmetic

Two steps of the recurrence already show the finite data we need:

$$X_{\ell+2} = b_{\ell+1} b_\ell X_\ell - (b_{\ell+1} m_\ell + m_{\ell+1}).$$

If BX_ℓ is an integer, this identity determines the residue of $BX_{\ell+2}$ modulo the product of the two bases. We will compare the least positive representative of that residue with the upper bound for the endpoint tail.

For a window starting at $\ell \geq 0$ and having $h \geq 0$ steps, define its product of bases and accumulated numerator from (5) by

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \quad W_{\ell,h+1} = b_{\ell+h} W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h} F_{\ell,h} + m_{\ell+h}.$$

Induction on h gives

$$X_{\ell+h} = W_{\ell,h} X_\ell - F_{\ell,h}. \tag{7}$$

For integers $W \geq 1$ and t , define $\text{lpr}_W(t) = 1 + ((t - 1) \bmod W)$, using a remainder in $\{0, \dots, W - 1\}$. Thus a zero residue is represented by W , not 0. Throughout this section use the upper bound

$$K(B, a) = 90B(a + 1)^2. \tag{8}$$

Lemma 5.1 (least positive residues). *If d is a positive integer with $d \leq K$ and $d \equiv -BF \pmod{W}$, where $W \geq 1$, then $\text{lpr}_W(-BF) \leq K$.*

Lean: [paper finite endpoint obstruction](#).

Proof. Every positive representative of the residue is at least its least positive representative, so $\text{lpr}_W(-BF) \leq d \leq K$. $\square$

For example, the window starting at $\ell = 1$ with length 6 has $W_{1,6} = 648000$ and $F_{1,6} = 524431$. At $B = 1$ its least positive residue is 123569, exceeding $K(1,7) = 5760$. This rules out an integral X_1 , but not a rational denominator that would clear only at a later index. The theorem therefore asks for windows beyond every prescribed starting index, as well as for every eligible denominator.

Theorem 5.2 (a residue criterion for irrationality). *The number S is irrational if and only if*

$$\begin{aligned} &\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ &\text{there are } \ell \geq a_0, h \geq 1 \text{ such that } \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell + h). \end{aligned} \quad (9)$$

Lean: [short window equivalence](#).

Proof. Assume (9) and suppose S is rational. Theorem 4.6 supplies a positive integral carry $d_a = BX_a$ after its onset. Choose a window in (9) beyond that onset. Equation (7) gives $d_{\ell+h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}$, contradicting Lemma 5.1 and the upper bound.

Conversely, let S be irrational and fix $B, \ell \geq 1$. Equation (6) makes BX_ℓ nonintegral. Put $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0, 1)$. Since $W_{\ell,h} \geq 2^h$ and $X_{\ell+h} = O((\ell + h + 1)^2)$, for all sufficiently large h the number $\delta + BX_{\ell+h}/W_{\ell,h}$ lies in $(0, 1)$. The window identity shows that the integer $\lceil BX_\ell \rceil W_{\ell,h} - BF_{\ell,h}$ lies strictly between 0 and $W_{\ell,h}$ and is congruent to $-BF_{\ell,h}$. Hence

$$\text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its exponentially growing first term eventually exceeds $K(B, \ell + h)$. This proves (9), in fact from each fixed start. $\square$

The proved bound $90B(a+1)^2$ controls every positive integral BX_a that rationality would force. Polynomial growth alone does not suffice: the zero bound would make the residue inequality automatic. The companion, Section 7, treats smaller and more general bounds. No finite search establishes the unbounded quantifiers in (9).

The remaining task has the following three equivalent formulations.

Problem 5.3 (the repeated three-prime target). Prove that $S = \mathcal{R}_{\{2,3,5\}}$ is irrational.

Problem 5.4 (no reduced tail is an integer). Prove that for every $a \geq 1$ and every integer $B \geq 1$ coprime to 30,

$$BX_a \notin \mathbb{Z}. \quad (10)$$

Problem 5.5 (residue inequalities after every starting index). Prove (9).

Indeed, one integral scaled tail makes S rational by (6), whereas denominator clearing gives such tails from a rational S . Theorem 5.2 supplies the third equivalence. These arithmetic assertions remain unproved. The companion's finite certificate (Section 8) and comparisons with other approaches (Section 9) are supplementary, not premises of this argument.

Statements and declarations

Proof sources. The two-prime identities and their transcendence conclusion are formalised in `TwoPrimeSums.lean`. The latter theorem retains the cited Bugeaud–Laurent transcendence theorem as an explicit hypothesis; that external theorem is not formalised here. Fan’s priority is retained from the supplied forum record; the live thread and catalogue could not be rechecked for this revision. The formal sources for the rank theorem and the tail recurrence are listed below. The supplied `LEAN_INDEX.json` identifies the public snapshot as `6b78209ab63a8c643281115f8628a3be79ff7ec7` and marks selected declarations `ci_checked`. Its recorded successful compilation is at an earlier commit; the build step in the pinned run was skipped. This revision reports that evidence, not a fresh compilation of the snapshot. The separate Palomar release at `52f29ad1` selects the arbitrary-order uniform-minor and non-separation statements, not just the 2×2 example.

The finite cut-rank proposition is the checked declaration [finite cut rank](#). Its column-range hypothesis permits repeated columns and arbitrary column order. The modular-minor corollary is the checked declaration [uniform modular minors](#). There the injective maps are chosen before the universal quantifiers over B and k , and the modulus is assumed at least 2 and coprime to 30. Both modules compiled under Lean 4.29.1 during this source migration. The exact-denominator corollary has an ordinary proof above. The moving-boundary example is proved in the companion, without a new formalisation claim. No Comparator, Isabelle or NanoDa run was performed for this exposition revision. Neither the listed checks nor the finite calculations certify every sentence of the papers, and none supplies the required windows for every eligible denominator and arbitrarily late starts.

Artefact and data availability. The source links retain their original revisions, which can differ from the snapshot in the supplied index. A link identifies a declaration; it does not establish that declaration’s build status at another commit. The companion contains the detailed bounds and supplementary arguments; its source inventory explains the index and normalisation conventions. AI agents carried out most of the research and drafting. The work has not had independent human mathematical review.

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A Guide to the formal sources

There are two separate arguments. The rank proof uses the height formula, row and column rescaling, and a threshold determinant. The arithmetic argument uses the shell

coefficients, the tail recurrence, denominator clearing and least positive residues. The table identifies the relevant Lean modules. Declaration names appear only in this source guide and the index that follows; the links preserve the revisions used in the original note.

Result	Source and scope
Kernel rank	KernelCarryRank: arbitrary-order uniform minors and no finite rational separation; also selected in the Palomar release.
Tail recurrence	DyadicShellSummability, RationalityCarryBridge: summable tails and denominator clearing, not merely an abstract recurrence.
Quadratic bounds	PaperR8RankMajorant and ActualSharpTailMajorantR10: the companion's two bounds in the sum of the boundary exponents, the smaller using the order of the jumps.
Residue equivalence	CofinalWindowEscapeEquivalence and R12/OcticWindowBand: the residue contradiction and equivalence with irrationality, not a proof of the required unbounded window existence.
External input	The Lean two-prime transcendence theorem assumes the cited Bugeaud–Laurent result (including the Loxton–van der Poorten case); it does not formalise that external theorem.

The public index supplied for this revision identifies commit 6b78209ab63a8c643281115f8628a3be79ff7ec7. The links below still use the note's original commit, not that newer snapshot. A source location, a successful build, a selected Comparator statement and an axiom audit establish different things; none substitutes for the others.

Scope of the supplementary literature. The companion develops the comparisons omitted from the main argument. Its Section 5 relates the bounded-ratio argument to Erdős–Taylor [10, Theorem 1, p. 600] and Fan [11, Lemma 3.1, p. 7]; Section 6 compares the exact carry restrictions and polynomial-radix assumptions of Hančl–Tijdeman [12, Theorems 2.2, 3.1 and 4.2]. The Isabelle development of Koutsoukou–Argyryaki and Li [18] verifies classical criteria, not the present series.

Section 9 of the companion separates the finite-difference hypotheses of Luca–Ouaknine–Worrell [13, Definition 5, Theorems 6 and 8, Claim 10] from quadratic growth alone, and checks the obstacles to using the echoing [14], digit-complexity [19, Theorem 1] and multivariate Mahler [20] results. These comparisons do not establish the missing hypotheses for S .

For broader background, the companion distinguishes work on fixed-prime semi-groups [15, 16] from other unit-fraction irrationality problems [17]. No estimate from those papers is used in this note's counting argument.

B Pinned Lean sources

The following index retains every original declaration link. Line numbers refer to the unchanged commit named by `\commit`, not to the newer source tree supplied with this revision.

- smooth prefix lcm eq three prime height
- smooth prefix lcm eq of same log cell
- smooth prefix lcm first log step
- smooth prefix lcm second log step
- smooth prefix lcm third log step
- three prime jump set with origin card
- finite smooth kernel sum grouped by height
- dyadic block base235 cases
- dyadic internal power exponent unique
- smooth exponent shell card quadratic
- smooth3 val
- three prime height
- three prime kernel q
- smooth prefix exponents
- smooth prefix lcm
- smooth3 val dvd three prime height of mem
- smooth prefix lcm dvd three prime height
- pure first mem smooth prefix exponents
- pure second mem smooth prefix exponents
- pure third mem smooth prefix exponents
- same three prime log cell
- three prime height eq of same log cell
- three prime kernel q eq of same log cell
- positive prime powers
- positive prime powers card
- one not mem positive prime powers
- positive prime powers disjoint
- three prime positive jump set
- three prime positive jump set card
- three prime jump set with origin
- three prime height first log step

- three prime height second log step
- three prime height third log step
- smooth exponent box
- smooth point height
- smooth height fiber
- smooth height fiber kernel sum
- three prime height le cube
- kernel 235 origin
- kernel 235 two
- kernel 235 three
- kernel 235 six
- kernel 235 not rank one
- tail state step
- tail state step eq
- smooth exponent shell
- exponent unique in short interval
- smooth exponent shell card le drop first
- smooth exponent shell card le drop third
- sorted pair quadratic
- dyadic internal power
- dyadic block base235 mem interval
- least positive residue
- least positive residue pos le
- least positive residue mod eq
- residue escapes window
- no bounded positive state of residue escape
- residue le bound of bounded positive state
- no bounded positive int state of least positive residue
- window base
- window forcing
- affine recurrence window

- window forcing const mul
- integral carry window
- smooth3 val dvd three prime height of le
- integral carry cancel common factor
- reduced carry pos le of common factor
- reduced integral carry window
- cofinal local window escape
- no positive reduced carry of cofinal local window escape
- kernel 235 minor eq neg one fifteen
- least positive residue eq nat abs of pos le mod eq
- bounded radix zero or cofinal far
- least positive residue window forcing eq carry
- carry eq residue digit add coboundary
- cartier realisation
- finite realised span of factorisation
- carry residue mem interval
- residue digit mem interval
- channel block null iff channel potential
- no carry lift of error bound below two pow
- no bounded carry lift of block null two anchors
- binary lift two anchors force first block sum one
- no unit accurate lift with two anchors and first two block null

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